

INDUSTRY REPORT

# Electric Vehicle (EV) Infrastructure

## Global Market Analysis & Outlook

2025–2026 Edition | Based on IEA Global EV Outlook 2026 & Latest Market Data

|                                      |                                        |                                     |                                            |
|--------------------------------------|----------------------------------------|-------------------------------------|--------------------------------------------|
| <b>~\$40–48B</b><br>2025 Market Size | <b>\$383–493B</b><br>Projected by 2035 | <b>~22%</b><br>India CAGR (2026–35) | <b>20.7M units</b><br>Global EV Sales 2025 |
|--------------------------------------|----------------------------------------|-------------------------------------|--------------------------------------------|

Prepared May 2026 | Sources: IEA Global EV Outlook 2026, Grand View Research, Precedence Research, IBEF, PM E-DRIVE Scheme Data

# 1. Market Overview

EV infrastructure encompasses the structures, machinery, and equipment necessary to support electric vehicles — including battery chargers, rapid chargers, and battery exchange stations. The ecosystem spans hardware manufacturing, network software, grid integration, and end-user services. In 2025, the market crossed a pivotal threshold: global EV sales exceeded 20 million units for the first time, representing over 25% of all new cars sold worldwide.

## 1.1 Global Market Size & Revised Forecasts

Multiple research firms place the 2025 global EV charging infrastructure market between \$40 billion and \$48 billion depending on scope. Forecasts to 2035 have been revised upward significantly, ranging from \$238 billion (Grand View Research, 25% CAGR) to \$493 billion (Precedence Research, 26.3% CAGR). The wide range reflects differences in whether analysts count equipment only or include network services, energy aggregation, and software.

| Segment                            | 2025 Estimate                    | 2035 Projection   | CAGR             |
|------------------------------------|----------------------------------|-------------------|------------------|
| Equipment & Installation (narrow)  | ~\$40–42B                        | ~\$238B           | ~25%             |
| Broad Infra incl. Network Services | ~\$47–48B                        | ~\$383–493B       | 25–26%           |
| DC Fast Charger Revenue Share      | ~70–73% of infra revenue         | Growing           | Dominant segment |
| India EV Charging Market           | ~\$350–500M                      | \$113B+ ecosystem | ~22% (stations)  |
| V2G Market                         | Nascent — commercial pilots 2025 | ~\$47B by 2032    | High growth      |

## 1.2 Regional Breakdown

- **Asia-Pacific:** Dominates with 53–68% global market share in 2025 (varies by methodology). China alone holds ~65% of global public charging stock and ~60% of the light-duty EV fleet.
- **China specifics:** Over 80% of global fast-charger stock is in China. Fast charger count surged from 1.2 million (2023) to 1.6 million (2024). EVs reached nearly 55% of all car sales in 2025.
- **Europe:** Crossed 1 million public charging points in 2024 (35%+ YoY growth). Fastest-growing major region by CAGR, driven by 2035 ICE ban timelines.
- **North America:** US had 11,687 public fast-charging stations and ~60,000 ports as of mid-2025. Tesla still leads with 54.3% all-time DC fast-charging share; Electrify America second at 7.9%.
- **India:** ~27,000–29,000 public stations as of early 2026, spread across 849 cities. Karnataka leads with ~6,000+ stations, followed by Maharashtra, Uttar Pradesh, and Tamil Nadu.

#### **Key Insight: Fast Charger Dominance**

*DC fast chargers account for roughly 70–73% of global EV infrastructure revenue due to high installation, hardware, and maintenance costs. The fast charger segment is also the fastest-growing, with the first 1,000-volt EV models entering the market in 2025 and sub-10-minute charging emerging in 2026.*

### **1.3 Vehicle-to-Grid (V2G): From Concept to Commercial Reality**

V2G moved from pilot to commercial deployment in 2025. Smart charging systems can reduce peak electricity demand by shifting charging loads, while V2G allows EVs to feed energy back to the grid. Commercial V2G services for private EV owners started appearing in 2025, though adoption remains limited due to fragmented regulations and a small number of compatible vehicle models. Individual V2G participants can earn several hundred to \$1,000 annually. The broader V2G market is projected to reach \$47 billion by 2032.

## 2. Competitive Analysis

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The global EV infrastructure market is fiercely competitive, with players differentiating on charging speed, network ubiquity, software interoperability, and grid integration. The US market in particular is entering a 'Charging 2.0' phase — large retailers, automakers, and energy companies are now building thousands of stations where the primary business is not charging itself.

### 2.1 Major Global Players

#### Tesla

- **Market Position:** Largest DC fast-charger operator globally, with 45,000+ Superchargers and 54.3% all-time share of US DC fast-charging ports.
- **Competitive Edge:** Proprietary network with unmatched reliability and seamless plug-and-charge. The NACS standard has been adopted by virtually all major automakers, extending Tesla's ecosystem.
- **2025 Development:** Average Supercharger site now has 15.1 ports vs 3.8 for non-Tesla stations, signalling a network design advantage.

#### ChargePoint

- **Market Position:** Leads Level 2 (AC) commercial and residential charging with 250,000+ activated ports across North America and Europe.
- **Competitive Edge:** Asset-light, software-driven model — a cloud-based management system sold to third-party station owners, fleets, and retailers. Among the top 3 DC fast-charging networks in the US.

#### EVgo

- **Market Position:** One of the largest US public fast-charging networks, within 10 miles of 40% of the US population.
- **Competitive Edge:** Retail hub placement strategy, 100% renewable energy, and fleet partnerships with GM and Pilot Travel Centers.

#### bp pulse

- **Market Position:** A leading 'Charging 2.0' entrant — 29,000 charge points globally as of 2025, targeting 100,000+ worldwide by 2030.
- **Competitive Edge:** Leverages bp's network of ~8,500 US retail sites. Focused on ultra-fast airport and metro hub charging. \$1B committed to US EV infrastructure by 2030.
- **2025 Development:** Opened sixth major US airport charging hub (Chicago). Introduced Charging-as-a-Service (CaaS) model to reduce fleet operator upfront costs.

#### Electrify America

- **Market Position:** Second-largest US DC fast-charging network at 7.9% market share. Volkswagen-backed with a mandate to build out US charging access.
- **Competitive Edge:** Focus on high-power 150–350 kW chargers along interstate corridors. ABB hardware powers a significant portion of its network.

### ABB E-mobility

- **Market Position:** Among the largest EVSE hardware manufacturers globally. B2B focus — powers ChargePoint, Electrify America, and others.
- **Competitive Edge:** Multi-standard (CCS, NACS, CHAdeMO) compatibility, ultra-fast Terra 360 chargers, and growing presence in megawatt charging for heavy-duty trucks.

### State Grid Corporation of China

- **Market Position:** Dominant utility-backed operator in China, which accounts for 65% of global public charger stock.
- **Competitive Edge:** National utility integration at unmatched scale. China's State Grid is effectively the backbone of the world's largest EV infrastructure network.

### India — Tata Power, ChargeZone & Adani Evolve

- **Tata Power EZ Charge:** Largest listed EV charging network in India with 3,500+ stations. First to span all 4 major metro cities. Partners with Tata Motors, MG Motor, and JLR. App-based RFID and dynamic pricing.
- **ChargeZone (TechSo):** Over 13,500 charging points across India and the UAE. In Nov 2025 launched Project E-DHARA — targeting 100% renewable-powered charging via 100 highway solar hubs by 2026.
- **Adani Evolve:** Installing ultra-fast 120–180 kW DC chargers along national highway corridors. Targeting 10,000+ charge points by 2030, integrated with food plazas and Adani road networks.
- **Ather Grid:** 2,700+ fast-charging points in 230+ cities, primarily for two-wheelers. Moving toward open interoperability via India's LECCS standard.

## 2.2 Competitive Summary

| Company     | Primary Strength                   | Geography          | Key 2025–26 Move        |
|-------------|------------------------------------|--------------------|-------------------------|
| Tesla       | Proprietary network, NACS adoption | Global, B2C        | 15+ port avg. site size |
| ChargePoint | Software platform, asset-light     | N. America, Europe | Top 3 US DCFC           |
| EVgo        | Retail hub strategy, renewables    | United States      | Fleet partnerships      |
| bp pulse    | Fuel retail integration, CaaS      | Global             | Airport hub expansion   |

| Company           | Primary Strength                    | Geography        | Key 2025–26 Move       |
|-------------------|-------------------------------------|------------------|------------------------|
| Electrify America | Interstate corridors, high-power    | United States    | VW ecosystem growth    |
| ABB E-mobility    | Hardware, multi-standard            | Global, B2B      | Megawatt charging      |
| State Grid China  | Utility integration, national scale | China / Asia-Pac | 80%+ global fast stock |
| Tata Power        | Metro coverage, app ecosystem       | India            | All 4 metros covered   |
| ChargeZone        | Renewable-powered highway hubs      | India, UAE       | Project E-DHARA        |
| Adani Evolve      | Highway DC fast charging            | India            | 10,000 pts target 2030 |

## 2.3 Emerging Competitive Trends

- **Charging 2.0 Players:** In the US, large non-charging businesses (Walmart, Mercedes-Benz HPC, bp pulse, Rivian) are scaling up rapidly. In 2025, 12 operators are expected to open 400+ ports each vs. just 4 in 2024.
- **Megawatt Charging for Freight:** ABB, BYD, and emerging Chinese players are pioneering megawatt-class chargers for electric trucks. In China, electric truck sales outnumbered LNG truck sales for the first time in 2025.
- **Renewable Integration:** Operators are competing on green credentials. ChargeZone's E-DHARA, EVgo's 100% renewable claim, and solar-integrated urban hubs in Mumbai and Bengaluru reflect this trend.
- **Software & Roaming:** Companies offering seamless cross-network payment (CaaS, eMSP roaming) are capturing higher user loyalty as fragmentation remains a top driver of range anxiety.

### 3. Trends & Drivers

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The EV infrastructure ecosystem is undergoing rapid transformation driven by technological breakthroughs, shifting government policy from incentives to mandates, and evolving consumer expectations.

#### 3.1 Technological Advancements

- **Ultra-Fast & 1,000-Volt Charging:** First 1,000-volt EV models entered the market in 2025. Sub-10-minute charging times emerged in 2026. New US stations average 250–400 kW power delivery vs. 150 kW in earlier deployments.
- **Smart Grids & V2G:** Smart charging (V1G) shifts loads to reduce peak demand. V2G (bi-directional) entered commercial rollout for private owners in 2025, though fragmented regulations limit scale.
- **Battery Innovations:** Solid-state batteries, improved thermal management, and lithium-ion pack prices below \$100/kWh (achieved second consecutive year in 2025 per BloombergNEF) are reducing total EV costs and improving charge cycle efficiency.
- **Battery Swapping:** Particularly relevant in India for two- and three-wheelers. Government mandating standardization for commercial fleet operations under PM E-DRIVE.

#### 3.2 Regulatory Changes

- **From Incentives to Mandates:** India's PM E-DRIVE (Oct 2024–Mar 2026) allocated Rs 10,900 crore (\$1.3B), including Rs 2,000 crore specifically for 72,000+ public chargers — compared to FAME II's 9,300 chargers over 5 years.
- **China Tightening:** From 2026, China's full NEV purchase tax exemption is replaced by a 50% reduction linked to vehicle energy efficiency — reducing per-vehicle government support while sales continue growing.
- **US Federal Funding:** \$7.5B IIJA allocation continues to deploy, with state-level programs like ComEd's Make Ready Rebate accelerating private hub investment.
- **India Charger Spacing Mandates:** Policy mandates 1 charger per 3x3 km in cities and 1 per 25 km on highways. Maharashtra waives tolls for EVs and requires chargers every 25 km.
- **CAFE & CO2 Standards:** Stricter fuel economy and emissions targets continue forcing automakers to scale EV production and phase out ICE vehicles globally.

#### 3.3 Shifts in Consumer Behavior

- **TCO Breakeven Reached:** In India, two- and three-wheelers have already achieved TCO breakeven at low daily mileage thresholds, accelerating mass adoption from the bottom up.
- **EV Sales Milestone:** Global EV sales broke 25% of total new car sales in 2025 — a psychological threshold that signals mainstream adoption, not just early adopter demand.

- **Convenience Expectations:** With charging 2.0 infrastructure, users now expect minimum 10-stall stations with 250+ kW power, app-based slot booking, real-time availability, and seamless cross-network payments.
- **India's Segment Sequencing:** 2- and 3-wheelers dominate adoption (TCO parity achieved), followed by commercial fleets (e-buses, trucks), before mass market 4-wheeler transition.



## 4. Future Forecasts

### 4.1 Projected Market Growth

| Market / Region                        | Projection          | Timeframe     | Source              |
|----------------------------------------|---------------------|---------------|---------------------|
| Global EV Infrastructure Market        | \$238B–\$493B       | By 2035       | GVR / Precedence    |
| Global EV Infrastructure CAGR          | 25–27%              | 2026–2035     | Multiple analysts   |
| Global EV Fleet (all types excl. 2/3W) | 450M+ (5x growth)   | By 2035       | IEA CPS             |
| Global EV Sales Share                  | ~50% of new cars    | By 2030 (APS) | IEA                 |
| Public Chargers Globally (slow)        | 14 million          | By 2035       | IEA CPS             |
| Public Chargers (fast + ultra-fast)    | 10M+                | By 2035       | IEA CPS             |
| India EV Charging Stations (volume)    | 11.58 million units | By 2035       | Expert Mkt Research |
| India Public Charging Points           | ~375,000            | By 2030       | IEA STEPS           |
| India Charging Market CAGR             | ~22%                | 2026–2035     | Expert Mkt Research |
| V2G Global Market                      | ~\$47B              | By 2032       | Persistence MR      |

### 4.2 India-Specific Outlook

India's EV infrastructure has grown fourfold since 2023 — from ~6,586 to 29,277 public stations by November 2025 — driven by PM E-DRIVE and FAME II combined with private investment exceeding Rs 10,000 crore. Karnataka leads with ~6,000+ stations, followed by Maharashtra, Uttar Pradesh, and Tamil Nadu. PSU oil companies (IOCL, BPCL, HPCL) are converting fuel station networks: IOCL alone targets 10,000 EV charge points. Despite this progress, approximately 85% of existing chargers are slow AC units — the fast DC charging buildout is still in early stages outside major metros.

#### India Infrastructure Gap — 2026 Reality Check

*As of March 2026: 27,737 installed / 22,753 operational public charging points across India. IEA projects ~375,000 points needed by 2030 (STEPS). That requires 50,000 new chargers per year — roughly 30% more than the 2024 addition rate. The gap is vast, but PM E-DRIVE's 72,000-charger target in just 2 years shows the policy intent to close it fast.*

### 4.3 Potential Risks & Challenges

- **Grid Overloads:** High-speed DC fast chargers create severe localized grid stress. Substation upgrades require heavy capex. Battery-buffered solutions (ADS-TEC Energy) are emerging as a workaround.
- **Raw Material Volatility:** Lithium-ion pack prices fell below \$100/kWh in 2025, but lithium, cobalt, and nickel supply chains remain exposed to geopolitical disruption and price spikes.
- **Network Fragmentation:** Inconsistent connectors, payment systems, and reliability across operators continue to create range anxiety. Roaming standards and NACS adoption are improving but not yet universal.
- **Policy Transition Risk:** PM E-DRIVE expires March 2026. India's FAME III details remain unclear. Policy gaps between schemes historically cause temporary slowdowns in investment and adoption.
- **Slowing US Deployment Risk:** Political headwinds around IIJA funding and federal EV policy create uncertainty for US charging investment timelines.
- **Utilization Economics:** Many public chargers operate below breakeven utilization. Station operators face a 'chicken-and-egg' dilemma — infrastructure must precede demand, but funding requires demonstrated demand.

## 5. Supply Chain Structure

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The EV infrastructure supply chain spans raw material extraction through to the end-user charging experience. It is distinct from the EV vehicle supply chain — focused on grid-connected hardware, software, and service networks rather than drivetrains and batteries.

### 5.1 Upstream: Raw Materials & Components

- **Critical Minerals:** Copper (for cables and connectors), aluminum, steel, lithium (for buffer batteries), and rare earth metals underpin all hardware.
- **Component Manufacturing:** Refined into EVSE components — charging cables, CCS/NACS/CHAdeMO connectors, metering chips, transformers, and power electronics.
- **Supply Risk:** India's PLI scheme for Advanced Chemistry Cells (ACC) is localizing battery manufacturing, which accounts for 40–50% of EV cost — reducing import dependency.

### 5.2 Midstream: Assembly & Software

- **Hardware Assembly:** Delta, Siemens, ABB, Exicom, Servotek (India-focused), and BYD assemble complete Level 2 and DC fast charging units.
- **Energy Management Systems (EMS):** Software layer for load balancing, usage monitoring, payment processing, and grid interaction. Increasingly AI-driven.
- **Global Logistics:** Maritime and regional distribution networks deliver to installation hubs. India's domestic manufacturing push under PLI aims to reduce hardware import dependency.

### 5.3 Downstream: Installation & Operations

- **Site Planning & Permitting:** Contractors assess grid capacity, negotiate with DISCOMs (India) or utilities (US/EU), and secure municipal permits. Spacing mandates (India: 3x3 km in cities, 25 km on highways) now guide site selection.
- **Installation:** Civil engineers and electricians mount hardware, install transformers, and connect to grid. EV charging in India is de-licensed, enabling faster private sector participation.
- **Charge Point Operators (CPOs):** Own and manage physical stations — Tata Power, ChargeZone, EVgo, bp pulse, Electrify America.
- **E-mobility Service Providers (eMSPs):** Manage consumer billing, roaming, and app interfaces — often distinct from CPOs, enabling multi-network payment with a single account.

### 5.4 End-User: The EV Driver

- **Discovery & Access:** Drivers use apps (PlugShare, Tata Power EV, ChargeZone, Tesla app) to find, navigate to, and initiate charging. RFID and plug-and-charge reduce friction.

- **Cost:** Indian public charging rates range from Rs 3–12/kWh depending on city and operator. Delhi charges among the cheapest (Rs 3–5/kWh), Mumbai among the higher (Rs 5–12/kWh).
- **Maintenance:** Fleet managers and station attendants handle firmware updates and physical repairs. Network uptime is a key differentiator — Tata Power's Network Operations Centre (Mumbai) monitors reliability nationally.

| Stage           | Key Players                      | Focus Area                   | India Context                  |
|-----------------|----------------------------------|------------------------------|--------------------------------|
| Upstream        | Mining cos., metals refiners     | Copper, lithium, rare earths | PLI-ACC for local battery mfg  |
| Midstream       | ABB, Siemens, Exicom, Servotech  | EVSE assembly, EMS software  | Domestic mfg push under PLI    |
| Logistics       | Maritime & regional distributors | Hardware distribution        | Import vs local sourcing shift |
| Downstream CPO  | Tata Power, ChargeZone, Adani    | Station ops, grid connection | De-licensed; DISCOM interface  |
| Downstream eMSP | Bolt.Earth, Statiq, YoCharge     | Billing, roaming, app UX     | LECCS open standard emerging   |
| End-User        | EV drivers, fleet managers       | Charging experience          | App-based; RFID; 85% slow AC   |

## Appendix: Key Data Sources

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This report draws on the following primary and secondary sources, updated to May 2026:

- IEA Global EV Outlook 2026 — electric vehicle charging, outlook for electric mobility
- IEA Global EV Outlook 2025 — charging infrastructure data and India projections
- Grand View Research — EV Charging Infrastructure Market Report 2033
- Precedence Research — EV Charging Infrastructure Market Size & Growth (Mar 2026)
- NITI Aayog — Electric Vehicles in India Report 2025
- PM E-DRIVE Scheme — Ministry of Heavy Industries (India, Oct 2024–Mar 2026)
- IBEF — Electric Vehicle Industry in India (Dec 2025 data)
- Paren — US EV Fast Charging State of the Industry Report Q2 2025
- InsideEVs — US EV Charging Q2 2025 Deployment Data
- Expert Market Research — India EV Charging Market 2026–2035
- BloombergNEF — Lithium-Ion Battery Price Survey 2025
- India Data Report — State-wise EV Charging Stations 2025–26
- Bolt.Earth — Government Incentives for EV Charging Infrastructure in India

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*Disclaimer: This report is compiled from publicly available industry research and market intelligence. Figures and forecasts vary across research firms due to differing methodological scopes. Readers should verify critical data independently before making investment or strategic decisions.*